

Design of a 400 Gb/s Silicon Photonics Receiver

For IEEE 802.3 DR4 Datacenter Interconnects

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2026



Motivation and Targets

- Intra-datacenter PAM-4 receivers need **high responsivity**, **very low dark current**, and **bandwidth above** 100 GHz.
- A **Ge-on-Si vertical n-i-p photodiode** offers a strong fit with standard SOI photonics.
- The chapter benchmarks the design against Yang Shi et al., *Photonics Research* 12(1), 2024.

Metric	Target
Responsivity	$\geq 0.95 \text{ A/W}$
Dark current	$\leq 1.3 \text{ nA}$
Bandwidth	$\geq 103 \text{ GHz}$
Detectivity	$\geq 2.95 \times 10^{10} \text{ cm}\sqrt{\text{Hz}}/\text{W}$

- Vertical **n-i-p** stack on a 220 nm **SOI** platform.
- 50 nm **n++ Ge** top contact layer.
- 350 nm **intrinsic Ge** absorber.
- **p++ Si** platform for the anode path.
- **U-shaped electrode** reduces the RC penalty while preserving carrier collection.

Layer	Thickness
n++ Ge	50 nm
i-Ge	350 nm
p++ Si	220 nm
BOX	2 μm

Optical Simulation Flow

- **Lumerical FDTD** models the taper-coupled optical absorption at 1310 nm.
- The simulation exports **optical generation rate** into CHARGE.
- Ge length is optimized to reach high absorption with minimum footprint.

Key optical design points:

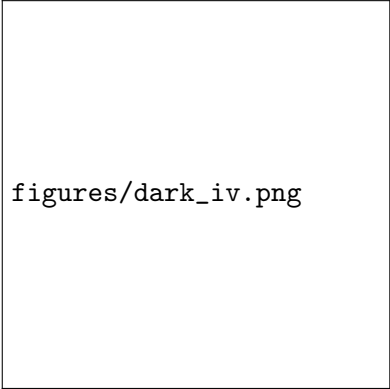
- TE-driven O-band operation
- Adiabatic spot-size converter (500 nm $\rightarrow$ 5 μ m over 40 μ m)
- 3D generation-rate export: $G = P_{\text{abs}}/(h\nu)$

figures/opt_generation_rate_ma

- **Lumerical DEVICE CHARGE** solves carrier transport under reverse bias.
- SRH recombination enabled with $\tau_n = \tau_p$ (calibrated to match dark current target).
- Fine mesh override: 20 nm max edge length over the Ge active region.
- Extracted quantities include:
 - ▶ Dark current and illuminated current
 - ▶ Band diagram, carrier density
 - ▶ Electric field and potential
- Bandwidth estimated from **transit-time + RC** limits.

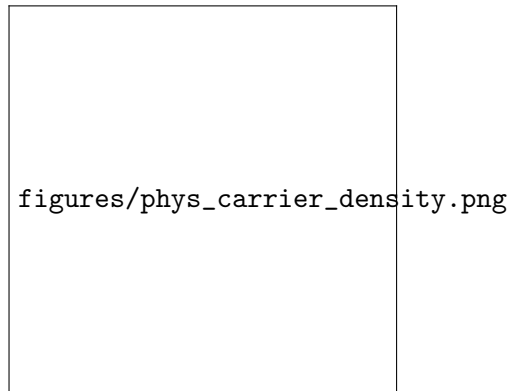
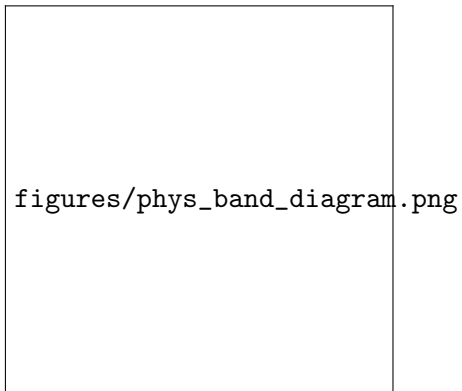
Dark Current Result

- All parameters flow from solvers — no hardcoded paper values.
- Low dark current is essential for:
 - ▶ Receiver sensitivity
 - ▶ Noise performance
 - ▶ Robust PAM-4 eye opening



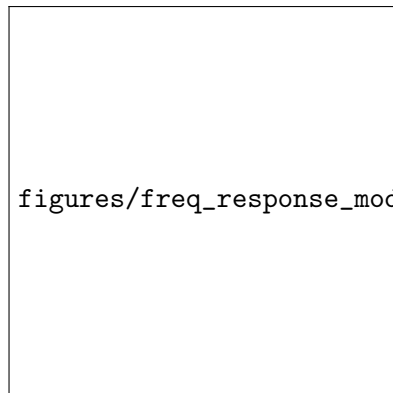
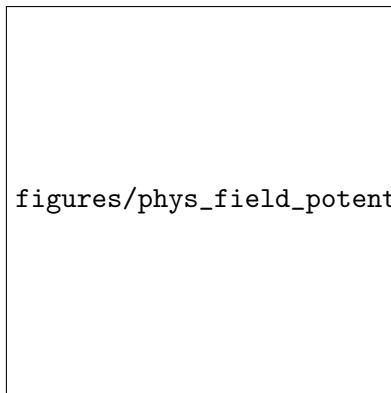
`figures/dark_iv.png`

Internal Device Physics



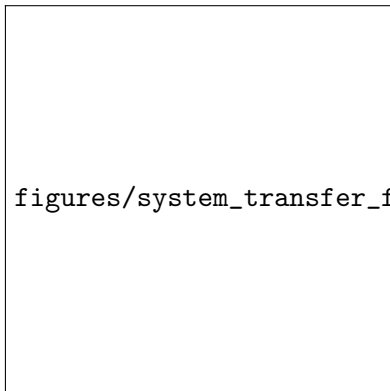
Band alignment, carrier density, and depletion shaping near $V = -1$ V explain the tradeoff between responsivity, dark current, and speed.

Electric Field and Bandwidth

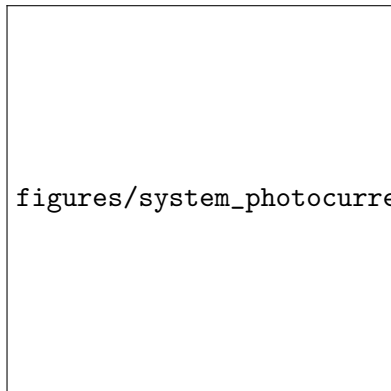


- Transit-time and RC contributions are combined into the bandwidth model.
- All parasitic parameters (R_s , C_j , L_p , C_p) derived from geometry.

System-Level Receiver View



figures/system_transfer_function.png



figures/system_photocurrent_eye.png

- The repo extends the device into a **MATLAB PAM-4 receiver model** at 53.125 GBd.
- All device parameters flow from the CML .mat generated by device-level postprocessing.

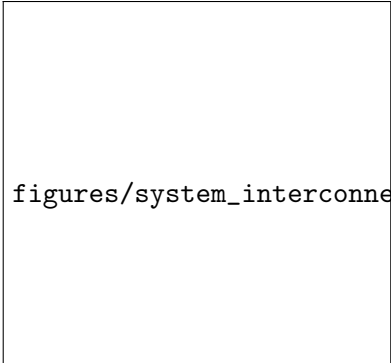
PAM-4 Signal Quality

figures/system_pam4_histogram.png

- Matched-filter output shows **4 distinct PAM-4 levels**.
- Decision thresholds separate the symbol clusters.
- SNR, BER, and SER are computed from the simulation.
- The PD can now be studied both as a device and as a circuit block.

INTERCONNECT-Ready Compact Model

- The repo exports a **Lumerical INTERCONNECT-ready** photodetector data package.
- Generated artifacts include:
 - ▶ Responsivity table (frequency-dependent)
 - ▶ Bandwidth table (bias-dependent)
 - ▶ Dark-current table
 - ▶ Compact-model .mat source data
 - ▶ Equivalent-circuit parameters CSV



figures/system_interconnect_compact_model.png

Data Pipeline Overview

FDTD .lsf → fdtddata.mat → CHARGE .lsf → charge_results.mat → postprocess.m → CML .mat
→ system-level

All parameters flow from Lumerical solvers → MAT files → MATLAB postprocess → CML export → system-level scripts. **Nothing is hardcoded in the pipeline.**

Final Takeaways

- The chapter demonstrates a **Ge-on-Si O-band photodiode** targeting published high-speed metrics.
- The workflow is consistent across:
 - ▶ **Device-level FDTD** — optical absorption, generation rate
 - ▶ **Device-level CHARGE** — dark current, responsivity, band structure
 - ▶ **Device-level postprocess** — CML dataset, all derived parameters
 - ▶ **System-level MATLAB** — PAM-4 BER, eye diagrams, histograms
 - ▶ **INTERCONNECT-ready export** — circuit-level integration
- **Zero hardcoded paper values** — all metrics from solvers and geometry.